

Weld seam extraction and path generation for robotic welding of steel structures based on 3D vision

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ABSTRACT

Recent advances in computer vision have provided new solutions for intelligent welding. However, existing vision-based weld seam extraction techniques exhibit limited adaptability to various workpieces in unstructured environments. Therefore, this paper proposes a three-dimensional vision-based method tailored for weld seam extraction and path generation. The proposed method synergizes a deep learning-based point cloud segmentation technique with an improved multi-scale point cloud registration algorithm to reconstruct the complete point cloud model of all weld regions in the workpieces. Subsequently, the welding paths and torch poses are calculated using an optimized multi-plane fitting algorithm integrated with geometry model of weld seam. Experimental validation on four workpieces demonstrates that the proposed method achieves good accuracy and outperforms the existing techniques in terms of efficiency and applicability, offering a robust solution for automated welding of steel structures.

1. Introduction

In recent years, with the rapid advancement of industrial automation and robotics, robotic welding has become a widely implemented application [1–6]. Traditionally, robotic welding operations have primarily relied on teach-and-playback mode and offline programming [7,8]. Although the teach-and-playback mode is operationally straightforward, it proves inefficient for complex workpieces and remains highly sensitive to assembly deviations [9]. Moreover, the accuracy of welding path is constrained by the proficiency of operator, making it challenging to meet high-precision welding requirements [10]. While offline programming facilitates a higher level of automation, substantial discrepancies frequently exist between the design model and the physical workpiece due to machining and fixture errors, leading to the failure of the generated welding paths [11].

The integration of computer vision technology has paved new avenues for intelligent welding [12,13]. Among existing studies, two-dimensional (2D) vision methods have been extensively adopted due to their hardware simplicity and ease of implementation. These

approaches apply image processing techniques to extract information, primarily for weld seam type recognition [14], molten pool monitoring [15], and welding quality inspection [16]. However, such methods suffer from restricted detection accuracy and inability to capture depth information, making them unsuitable for detecting three-dimensional (3D) features of complex weld seams. In contrast, 3D vision sensors provide superior spatial perception, making them highly effective for weld seam extraction and path generation [17–20]. Common 3D vision sensors mainly comprise laser structured light (LSL) sensors, binocular stereo vision sensors, RGB-D cameras, and coded structured light (CSL) sensors.

LSL sensors, by projecting laser line and integrating vision sensor, facilitate the 3D reconstruction of weld seams, offering notable advantages in real-time performance and recognition accuracy [21–25]. Wang et al. [26] developed a solution utilizing a single-line LSL sensor to address three distinct welding tracking scenarios: short, spatial curve, and long welds. Similarly, Ma et al. [27] designed an LSL sensor for spatial circular welds and achieved robust real-time weld tracking through self-updating template matching and random sample consensus

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(RANSAC). Addressing material diversity, Wu et al. [28] introduced a feature extraction method applicable to various reflective materials such as carbon steel, stainless steel, and aluminum alloys. Lei et al. [29] further incorporated polarization vision to mitigate noise on highly reflective materials, thereby reducing algorithm complexity and enhancing image processing efficiency. To address the challenge of indistinct laser stripe deformation in narrow seams, Fan et al. [30] proposed an initial point alignment and weld tracking method using an LSL sensor augmented with auxiliary light-emitting diode (LED) illumination, which proved effective for both planar and curved narrow weld seams. Furthermore, Nguyen et al. [31] constructed a three-line LSL sensor and developed a deep learning-enhanced tracking approach for narrow-gap square-groove welds, yielding improved accuracy and efficiency. However, the local sensing nature of LSL sensors restricts their field of view to the immediate vicinity of the weld seam, precluding the capture of global workpiece geometry. This limitation renders such sensors unsuitable for global path generation, confining their application primarily to real-time tracking scenarios.

Binocular stereo vision sensors rely on the principle of binocular parallax, capturing a scene from distinct viewpoints using dual cameras. Depth information is then calculated through stereo matching algorithms, enabling the determination of the 3D welding path [32,33]. Ma et al. [34] developed WeldNet, a deep neural network capable of automatically classifying diverse weld seam types, including butt, lap, and fillet welds. In addition, this approach achieves rapid and cost-effective localization of the weld initiation point based on dual-position monocular vision. However, these sensors struggle in environments characterized by complex backgrounds or sparse surface textures, which hinder feature point matching and compromise reconstruction accuracy. RGB-D cameras, capable of simultaneously capturing color (RGB) and depth information, provide notable advantages in cost-effective 3D imaging and environmental perception. Capitalizing on this technology, Zhou et al. [35] reconstructed workpiece point cloud model and applied an intensity-based edge detection algorithm for welding path extraction, demonstrating robustness across various workpieces. Nevertheless, inherent limitations regarding measurement range and precision remain distinct challenges.

Coded structured light (CSL) sensors extract high-precision 3D information by projecting encoded fringe or grid patterns onto target surface [36]. Compared to alternative 3D vision sensors, CSL sensors excel in global information acquisition and reconstruction accuracy. Yang et al. [37] proposed a V-groove weld feature extraction method based on point-to-plane distance utilizing a CSL camera, offering a novel approach for automatic extraction of weld seams. Lu et al. [38] designed a compact monocular CSL sensor that combines gray code and phase-shift code rasters, successfully extracting the position and pose of curved weld seams. Geng et al. [39,40] utilized a CSL camera to obtain point clouds of impeller blades and three types of workpieces, facilitating efficient and accurate robotic path planning. Ma et al. [41] presented a framework that registers the design model with the scanned point cloud of the workpiece through non-rigid registration, generating welding paths for butt, lap, and fillet spatial curve seams. Despite these contributions, two significant limitations persist in the current studies. First, existing studies predominantly focus on workpieces with single orientation or specific structure, while research regarding complex structural components remains scarce. Second, most welding path generation strategies are tailored to controlled scenarios, lacking a unified method capable of accommodating diverse and unstructured welding environments. Consequently, the pivotal research question addresses: How can an automated and unified weld seam extraction and path generation framework be established for complex steel structures that ensures high accuracy and robustness independent of prior design models?

Table 1

Comparison of different 3D vision sensors.

Sensor type	Principle	Advantages	Limitations
LSL sensor	3D geometry reconstructed from the laser stripe deformation	High accuracy	Sensitive to reflectivity, local sensing
Binocular stereo vision sensor	Depth from two-view stereo matching	Flexible, low-cost	Reduced accuracy on reflective or textureless surfaces
RGB-D camera	Combines RGB with infrared structured light or time-of-flight	Fast, low-cost	Limited depth accuracy and stability
CSL sensor	Depth calculated by decoding the deformed encoded patterns	Robust & accurate in complex geometries	Higher complexity

To address this question, this study presents a weld seam extraction and path generation method for steel structures based on the CSL camera. The primary contributions of this study are summarized as follows:

- 1) A point cloud segmentation network is utilized to isolate the weld seam regions, improving robustness within complex industrial environments.
- 2) A multi-scale point cloud registration technique is introduced to reconstruct the complete point cloud model of all weld regions in the workpiece. This approach eliminates the dependence on prior design models, thereby enhancing adaptability of the method.
- 3) A real-time algorithm for multi-weld seam localization and welding pose generation is developed, advancing the automation level of the welding process.

This research provides a solution for intelligent welding of complex workpieces, offering both theoretical significance and practical application value. The remainder of this paper is organized as follows: [Section 2](#) describes the system configuration and the method framework. [Sections 3 and 4](#) present the acquiring of complete point clouds of the weld seam regions. [Sections 5 and 6](#) detail the automatic extraction of welding paths and poses. [Section 7](#) reports the experimental validation and discusses the performance results. [Section 8](#) provides the concluding remarks and future research directions.

2. Configuration and framework

Currently, the 3D vision sensors employed for robotic welding primarily include LSL sensors, binocular stereo vision sensors, RGB-D cameras, and CSL sensors. [Table 1](#) provides a comparative analysis of their operating principles, advantages, and limitations. The comparison reveals that CSL sensors exhibit a superior balance of robustness and high accuracy when dealing with complex geometries. Therefore, the CSL sensor is selected as the optimal visual perception device for this study.

2.1. System configuration

In this study, a CSL vision-based robotic welding platform was established, as illustrated in [Fig. 1](#). The platform consists of two main subsystems: the robotic system and the CSL vision system. The robotic system comprises a 6-axis manipulator, a controller, a welding torch, a welding machine, a wire feeder, and a shielding gas supply unit. Complementing this, the CSL vision system consists of a Mech-Eye NANO



Fig. 1. Welding robot system configuration.

Table 2
Parameters of CSL camera.

Parameters	Value
Working Principle	Coded Structured Light
Optimum Working Distance	300–600 mm
Field of View (FOV)	220 × 150 mm @ 0.3 m; 440 × 300 mm @ 0.6 m
Resolution	1280 × 1024
Repeat Accuracy	0.1 mm @ 0.5 m

camera and a processing computer. The CSL camera integrates a projector and an RGB camera. Detailed specifications are listed in Table 2. The computer is equipped with an Intel i5 processor and an NVIDIA 4060Ti graphics card with 16 GB of memory.

To establish an “eye-in-hand” configuration, the CSL camera is rigidly mounted on the end-effector of manipulator. Point cloud data of the workpiece is captured from multiple perspectives and transferred to

the computer through the TCP/IP protocol. Subsequently, the computer processes point clouds using algorithmic modules to generate welding paths and torch poses. Finally, the generated paths and poses are translated into robot script commands and sent to the robot controller through TCP/IP for execution.

2.2. Method framework

The framework of the proposed method is structured into four primary steps, as illustrated in Fig. 2:

- 1) The PointNet++ network is employed to segment point clouds acquired from multiple viewpoints, enabling the precise isolation of the weld seam region. This process involves data acquisition, augmentation, annotation, and network training.
- 2) Coarse registration of multi-view point clouds within weld seam region is executed utilizing tool pose data obtained from the robot

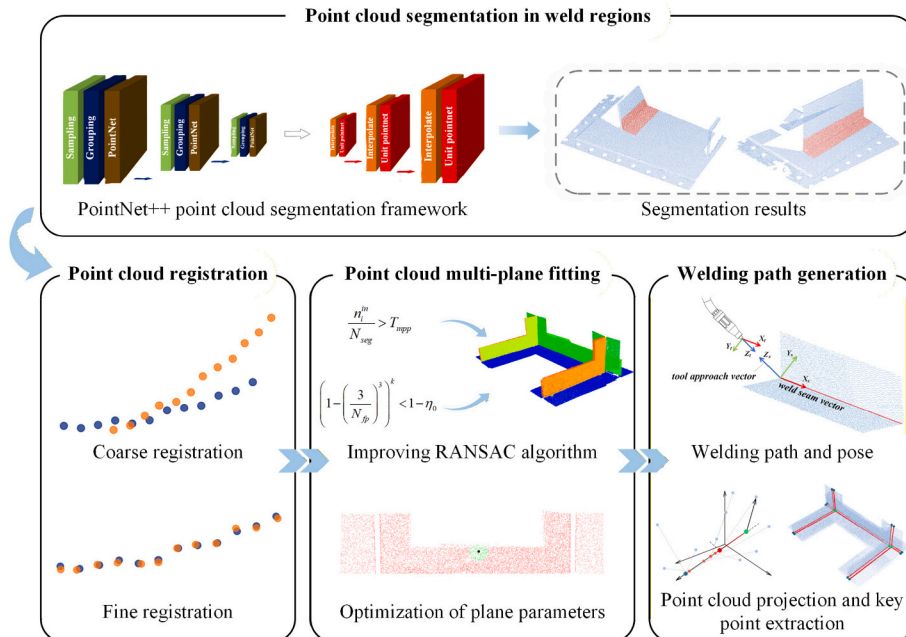


Fig. 2. Framework of weld seam extraction and path generation.

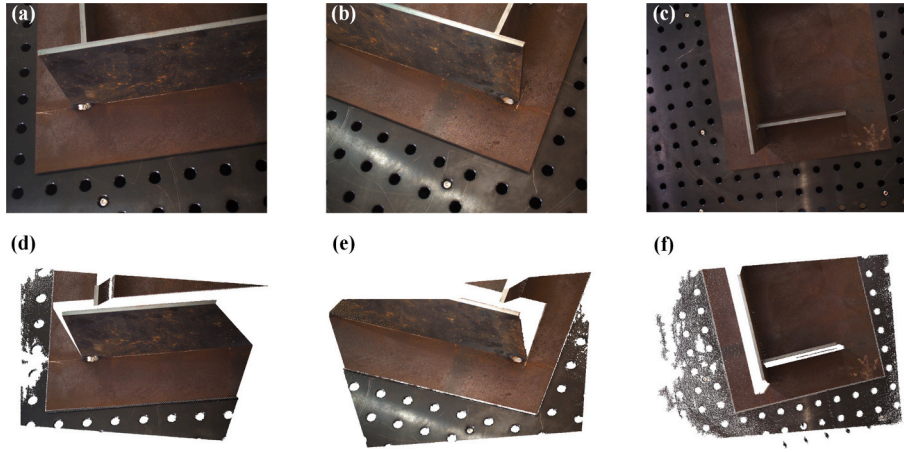


Fig. 3. Captured workpiece data: (a) ~ (c) RGB image; (d) ~ (f) point cloud.

controller and hand-eye calibration parameters. Subsequently, a proposed multi-scale iterative closest point (ICP) algorithm is applied to refine the registration results.

- 3) An improved RANSAC algorithm is deployed to fit multiple planes within the registered point cloud of the weld region. Precise plane parameters are extracted through plane separation and parameter optimization techniques.
- 4) The intersection lines of the fitted planes are computed to establish the welding direction vectors. By projecting plane inliers onto the weld vector, the start and end points of the weld are determined. Based on dihedral angle geometry, the tool approach vector is calculated to determine the optimal welding torch pose.

3. Segmentation of point cloud

The RGB images and point clouds of the workpiece were captured from multiple viewpoints using the CSL camera, as shown in Fig. 3.

To efficiently reduce the influence of noisy in point cloud data, this study employs deep learning techniques to perform segmentation of the weld seam regions. PointNet [42], the pioneering architecture capable of directly processing point cloud data, addresses the inherent unordered structure of point clouds using a symmetric MaxPooling operation. Additionally, to ensure invariance in rotation and translation, T-net subnetworks are incorporated to learn affine transformation matrices, aligning both the raw input and the extracted features. The network mapping function is defined as:

$$f(\{P_1, P_2, \dots, P_{N_d}\}) \approx g(h(P_1), h(P_2), \dots, h(P_{N_d})) \quad (1)$$

where P_m denotes the m -th input point cloud, N_d denotes the number of input point clouds, the function mapping is $f: 2^{\mathbb{R}^N} \rightarrow \mathbb{R}$, $h: \mathbb{R}^N \rightarrow \mathbb{R}^K$ is the per-point feature extraction function, $g: \underbrace{\mathbb{R}^K \times \dots \times \mathbb{R}^K}_n \rightarrow \mathbb{R}$ represents

the symmetric MaxPooling function.

However, PointNet exhibits limited capability to capture local features, which constrains its performance in complex scenes. To overcome this limitation, the authors introduced PointNet++ [43], whose segmentation architecture is shown in Fig. 4. The PointNet++ framework for point cloud segmentation includes two main components: feature extraction and feature propagation. Feature extraction is performed using a combination of sampling, grouping, and PointNet layers, facilitating the acquisition of high-dimensional, nonlinear representations. Feature propagation employs distance-based interpolation and skip connections to propagate hierarchical feature back to the original point set.

In this study, PointNet++ is used to segment the weld seam region from the point cloud of steel workpieces. First, the CSL sensor acquires data from varying poses, thereby enriching the dataset with various viewpoints. Subsequently, the dataset is expanded through data augmentation techniques such as rotation, random jittering, random translation, scaling, random sampling, and random point dropout. In addition, principal component analysis (PCA) is employed to compute normal vectors for each point. Finally, the weld seam neighborhood is annotated using dedicated software, with the region width maintained at approximately 40 mm. The PointNet++ model is trained on the expanded dataset to produce a trained model, which is used to extract the weld seam region. Furthermore, the segmented point cloud undergoes outlier removal to enhance data quality. The final segmentation results of weld seam region are depicted in Fig. 5.

4. Registration of point clouds with different viewpoints

Due to the limited FOV of camera and workpiece obstruction, single-view capture hinders the retrieval of complete point cloud models, as seen in Fig. 3(d)~(f). Therefore, this study introduces a multi-scale point cloud registration method for merging point clouds acquired from diverse viewpoints, as shown in Fig. 6. The method utilizes the manipulator pose data and hand-eye calibration parameters to achieve coarse registration, subsequently performing hierarchical fine registration based on overlapping regions.

In specific scenarios, as exemplified in Fig. 7(a) and (b), the segmented point clouds from adjacent viewpoints lack overlapping areas. Furthermore, involving more point clouds in the registration process improves overall accuracy. Therefore, multi-scale registration is performed on the original unsegmented point clouds. The derived transformation matrices are subsequently applied to the segmented point clouds.

4.1. Coarse registration of point cloud

To obtain comprehensive 3D point cloud of complex workpiece, a CSL camera mounted on a manipulator captures point clouds from various viewpoints. Before conducting hand-eye calibration, tool center point (TCP) calibration of the welding torch end is required. This study employs the four-point method to calibrate the torch tip coordinate system, as depicted in Fig. 8(a). A stationary reference p_f is selected within the robot workspace, and the torch tip is aligned with this point under four distinct poses. These poses are spatially distributed to enhance calibration robustness. During each contact, robot joint angles

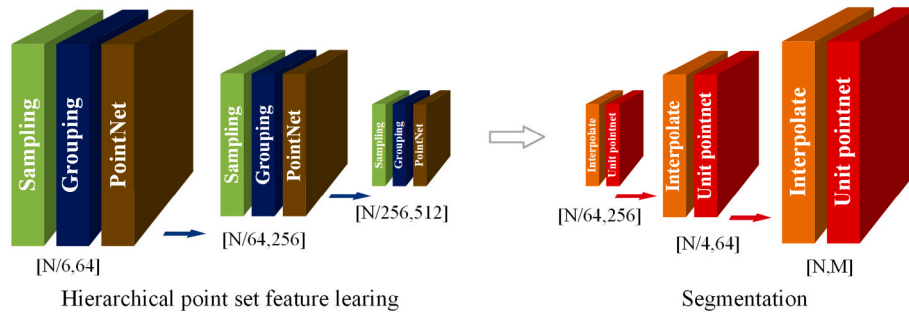


Fig. 4. PointNet++ point cloud segmentation framework.

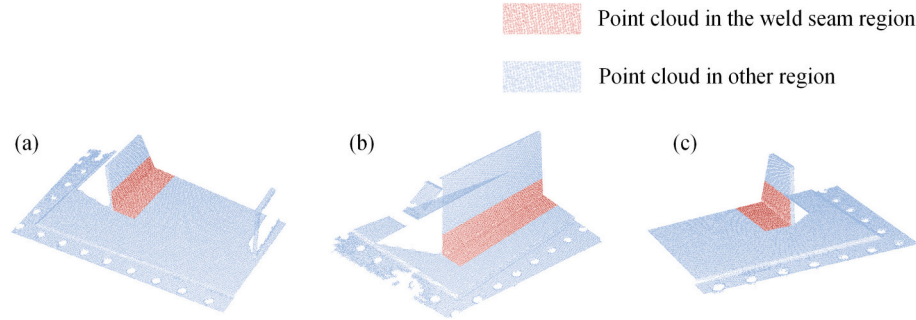


Fig. 5. Point cloud segmentation: (a) viewpoint 1; (b) viewpoint 2; (c) viewpoint 3.

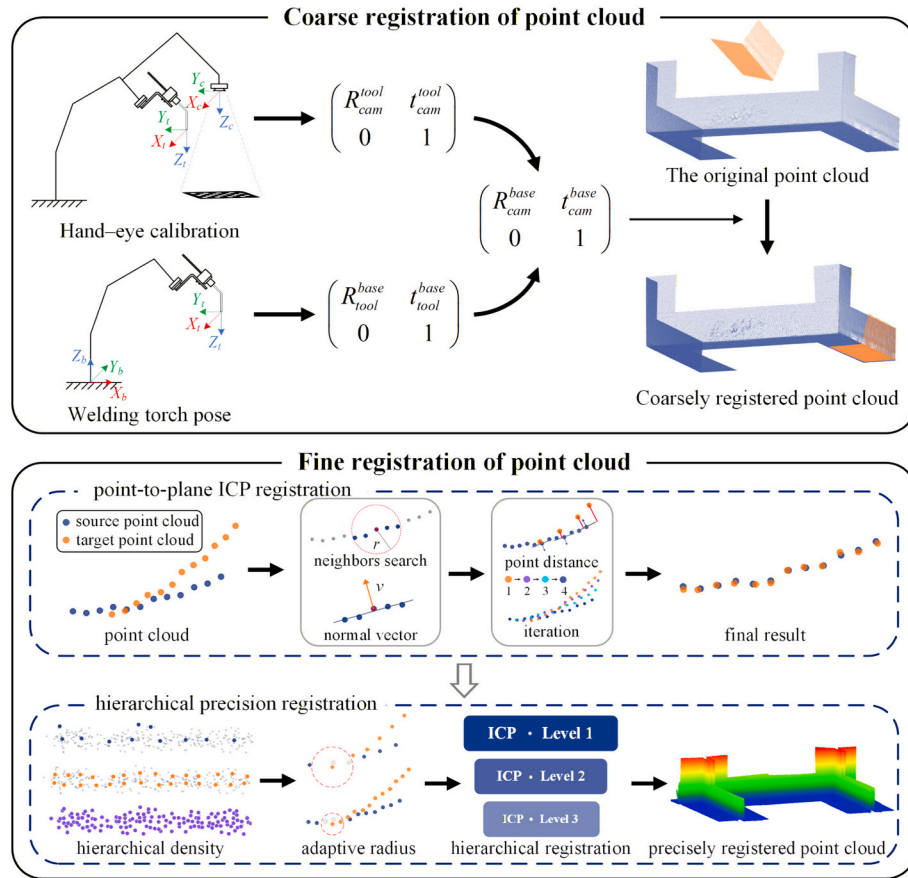


Fig. 6. Multi-scale point cloud registration.

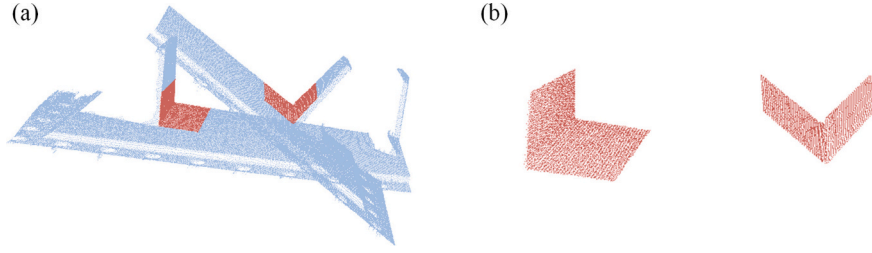


Fig. 7. Point cloud of neighboring viewpoint: (a) original; (b) segmented.

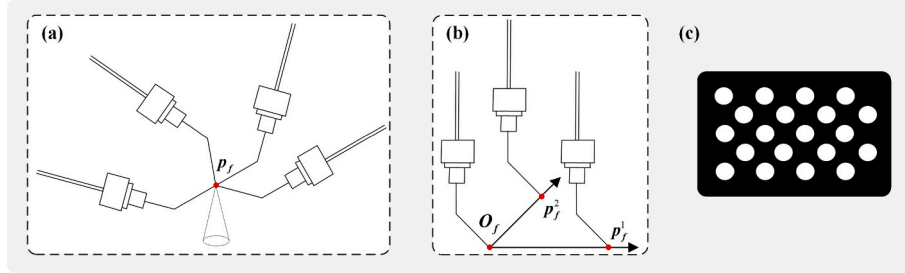


Fig. 8. TCP calibration and hand-eye calibration board: (a) TCP position calibration; (b) TCP orientation calibration; (c) hand-eye calibration board.

are recorded. By solving the geometric constraint equations using the least-squares method, the positional offset of the torch tip relative to the flange coordinate system is determined. To guarantee accuracy, a sharp torch tip is mandated to maintain contact consistency. Concurrently, robot joint configurations near kinematic singularities are strictly avoided to ensure calculation stability.

Subsequently, the orientation of the torch tip coordinate system is calibrated utilizing the designated X/Y-axis method, as illustrated in Fig. 8(b). The calibration procedure is executed as follows: first, a stationary reference point O_f within the workspace is chosen as the origin point for the torch tip coordinate system. The robot is manipulated to align the torch tip with this point, and the corresponding joint angle is recorded. Next, while maintaining a constant torch orientation, the tip is moved along the X-axis of the base coordinate system to the calibration point p_f^1 . This movement direction is defined as the positive X-axis of the torch tip coordinate system. After returning to the origin O_f , the tip is moved orthogonally to the X-axis to reach calibration point p_f^2 , thereby establishing the positive Y-axis. The distances from the origin O_f to points p_f^1 and p_f^2 are constrained within an optimal range, typically 50–100 mm. Crucially, the torch orientation must remain invariant throughout this process. By recording joint angles at each point, the homogeneous transformation matrices of the torch tip coordinate system relative to the base coordinate system are computed. Based on the geometric constraints, the transformation parameters relative to the flange coordinate system are derived, thus concluding the orientation calibration.

Following TCP calibration, a hand-eye calibration algorithm is employed to calculate the transformation matrix T_{cam}^{tool} from the camera coordinate system to the torch tip coordinate system. The calibration board depicted in Fig. 8(c), features a 5×4 circles with a radius of 20 mm. The robot is manipulated to acquire images of this board from 15 distinct viewpoints, which are then fed into the hand-eye calibration

algorithm for transformation matrix T_{cam}^{tool} .

At acquisition position of each point cloud, the manipulator provides a unique torch pose, consisting of a translation vector and Euler angles. The rotation matrices ${}_iR_{X_b}$, ${}_iR_{Y_b}$, and ${}_iR_{Z_b}$ about the X_b , Y_b , and Z_b axes are constructed from the Euler angles, and the overall rotation matrix is computed using Eq. (2). This matrix, combined with the translation vector ${}_i t_{tool}$, defines the transformation from the torch tip coordinate system to the base coordinate system, as formulated in Eq. (3).

$${}_iR_{tool}^{base} = {}_iR_{Z_b} \cdot {}_iR_{Y_b} \cdot {}_iR_{X_b} \quad (2)$$

$${}_i T_{tool}^{base} = \begin{bmatrix} {}_iR_{tool}^{base} & {}_i t_{tool} \\ 0 & 1 \end{bmatrix} \quad (3)$$

The point cloud is transformed according to its corresponding pose, as shown in Eq. (4). The transformed point clouds are combined to form a coarse registered point cloud. Subsequently, redundant points caused by overlapping areas are eliminated to improve computational efficiency. The final coarse registration result is shown in Fig. 9.

$$P_i^{base} = {}_i T_{tool}^{base} \times T_{cam}^{tool} \times P_i^{cam} \quad (4)$$

4.2. Fine registration of point cloud

Systematic errors in robot kinematics and calibration inaccuracies potentially reduce the precision of coarse registration, as shown in Fig. 9 (b). To address this issue, this study proposes a hierarchical fine registration method based on overlapping point cloud regions. This method progressively integrates higher-resolution point cloud details to optimize registration, as illustrated in Algorithm 1.

Algorithm 1. Improved multi-level ICP algorithm

Input:
 $P_{src} = \{P_{src}^1, P_{src}^2, \dots, P_{src}^m\}$: source point cloud, $P_{tgt} = \{P_{tgt}^1, P_{tgt}^2, \dots, P_{tgt}^m\}$: target point cloud

 $T_0 = \text{eye}(4)$: Initial transformation matrix

 n_s, S, T_d
Output: optimal transformation matrix T_{opt}
for i **from** 0 **to** $n_s - 1$ **do**:

 $P_{src} \leftarrow T_0 \cdot P_{src}$
 $P_{src}^i \leftarrow \text{VoxelSampling}(P_{src}, S[i]), P_{tgt}^i \leftarrow \text{VoxelSampling}(P_{tgt}, S[i])$
 $P_{src}^{ri} \leftarrow \text{RemoveStatisticalOutlier}(P_{src}^i), P_{tgt}^{ri} \leftarrow \text{RemoveStatisticalOutlier}(P_{tgt}^i)$
 $\text{ComputeNormals}(P_{src}^{ri}, P_{tgt}^{ri})$
for P_{src}^j **in** P_{src}^{ri} **:**
 $P_{tgt}^j \leftarrow \text{closest point to } P_{src}^j \text{ in } P_{tgt}^{ri} : d(P_{src}^j, P_{tgt}^j) = \frac{|Normal(P_{tgt}^j) \cdot (P_{src}^j - P_{tgt}^j)|}{\|Normal(P_{tgt}^j)\|}$
 $d_j \leftarrow |P_{src}^j - P_{tgt}^j|, \theta_j \leftarrow \angle(Normal(P_{src}^j), Normal(P_{tgt}^j))$
if $d_j < T_d(i)$ **and** $\theta_j < T_\theta$ **:**
 $\text{add } (P_{src}^j, P_{tgt}^j) \text{ to the matching pair of overlapping region set } \Omega_i$
end if
end for
 $T_{opt} \leftarrow \arg \min_T E(T) \text{ in } \Omega_i : E(T) = \sum_{j=1}^N \rho((T \cdot P_{src}^j - P_{tgt}^j) \cdot Normal(P_{tgt}^j))^2$
 $T_0 \leftarrow T_{opt}$
end for
return T_{opt}

(1) Multi-level voxel downsampling

Point clouds are divided into multiple voxel grids, and each voxel is represented by a single point to reduce data volume. Assuming the initial voxel size is v_0 , a sequence of n_s decreasing voxel scales is generated using the following formula:

$$v_i = v_0 + 0.1 \cdot (n_s - i), \quad i = 1, 2, \dots, n_s \quad (5)$$

where n_s is the number of scales in the multi-scale layer.

(2) Adaptive search distance

At each level i , the maximum search distance T_d for corresponding points is dynamically adjusted according to the voxel size:

$$T_d(i) = c_i \cdot v_i, \quad c_i \in \{3, 2.5, 2, 1.5, 1\} \quad (6)$$

where c_i is a predefined reduction factor controlling the search range when n_s is set to 5, which adapts dynamically according to specific rules as n_s varies. At coarser voxel scales, a larger search distance is required to capture correspondences under larger initial misalignments. Conversely, at finer voxel scales, the search radius needs to be gradually reduced to suppress mismatches and improve precision. The coefficient values were selected empirically through sensitivity analysis across multiple workpieces with varying geometric complexity. The results

confirmed that this setting achieves an optimal balance between registration stability and computational efficiency.

(3) Point cloud preprocessing

At each voxel level, both source point cloud P_{src} and target point cloud P_{tgt} are downsampled to generate sparse point sets P_{src}^i and P_{tgt}^i , respectively. The normal $\mathbf{n}_j$ of a point p_j is obtained by minimizing the eigenvalues of the covariance matrix:

$$\mathbf{n}_j = \arg \min_{\|\mathbf{n}\|=1} \mathbf{n}^T \mathbf{C}_j \mathbf{n} \quad (7)$$

where $\mathbf{C}_j$ is the covariance matrix defined by the k -nearest neighbors:

$$\mathbf{C}_j = \frac{1}{|N_j|} \sum_{p_k \in N_j} (p_k - \bar{p}_j) (p_k - \bar{p}_j)^T \quad (8)$$

where N_j is the neighbor set of p_j , $|N_j|$ is the number of neighbors, $\bar{p}_j$ is the centroid of the neighbor. Subsequently, statistical filtering is applied to remove outliers based on the criterion:

$$\frac{\|p_k - p_j\|}{\sigma_j} > \tau \quad (9)$$

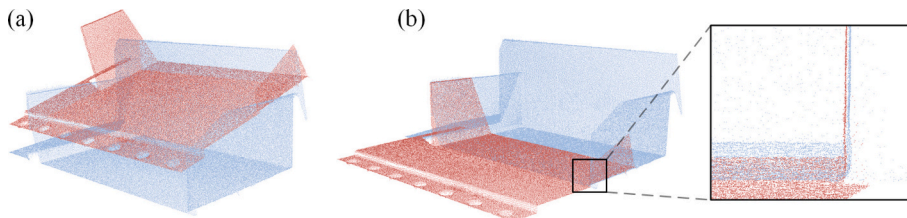


Fig. 9. Point cloud before and after coarse registration: (a) original point cloud; (b) coarsely registered point cloud.

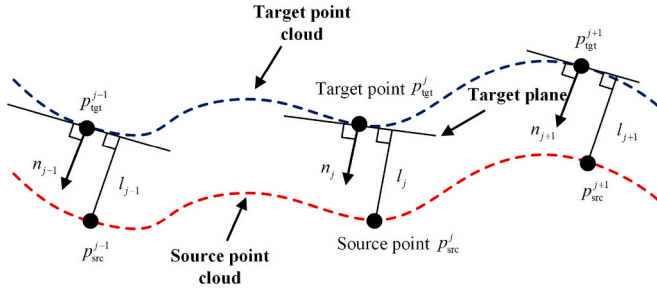


Fig. 10. Principle of the point-to-plane non-rigid ICP algorithm.

where σ_j is the standard deviation of the distances to neighboring points, and τ is a predefined threshold.

(4) Registration optimization

For each scale i , a KD-tree structure is used to search for the nearest neighbors between the preprocessed point sets P_{src}^i and P_{tgt}^i . Simultaneously, the Euclidean distances and normal angle deviations for all matching pairs are calculated. The final overlapping area is determined by adaptive search distance and angle threshold. To accommodate normal vector deviations induced by point cloud noise, an angular threshold of 45° is established. This threshold effectively balances filtering efficiency with matching accuracy. The optimization objective is formulated based on a point-to-plane non-rigid ICP formulation, as follows:

$$E(\mathbf{T}) = \sum_{j=1}^N \rho \left(\left((\mathbf{T} \cdot \mathbf{p}_{src}^j - \mathbf{p}_{tgt}^j) \cdot \mathbf{n}_j \right)^2 \right) \quad (10)$$

where $\mathbf{T} \in \mathbb{R}^{4 \times 4}$ is the transformation matrix, $\mathbf{p}_{src}^j$ and $\mathbf{p}_{tgt}^j$ are matched point pairs, $\mathbf{n}_j$ is the normal at $\mathbf{p}_{tgt}^j$, and ρ is the robust loss function (this study selects the L1 loss function: $\rho(x_i) = |x_i|$). The principle of the point-to-plane non-rigid ICP algorithm is shown in Fig. 10.

The optimization procedure proceeds as follows:

- 1) Initialize the transformation matrix $\mathbf{T}_0 = \mathbf{I}$ and apply it to the source point cloud.
- 2) Establish corresponding point set utilizing adaptive search distance and nearest neighbor search.
- 3) Minimize the objective function to update $\mathbf{T}_{m+1}$:

$$\mathbf{T}_{m+1} = \arg \min_{\mathbf{T}} E(\mathbf{T}) \quad (11)$$

- 4) Update the coordinates of the source point cloud:

$$\tilde{\mathbf{p}}_{src}^i = \mathbf{T}_{m+1} \cdot \mathbf{p}_{src}^i \quad (12)$$

Through these steps, point clouds from viewpoints 1 and 2 are registered. The convergence tolerance and maximum number of

iterations for the iterative process are provided by the Open3D library (Python version), which are set to 1×10^{-6} and 30, respectively. Subsequently, the registered point clouds from viewpoints 1 and 2 are merged, and the point cloud from viewpoint 3 is registered to the merged result. This incremental process is repeated for subsequent views, obtaining a complete unsegmented point cloud, as shown in Fig. 11(a). The rotation matrices obtained from the registration of unsegmented point clouds are applied to the segmented point clouds to reconstruct the complete point cloud in weld seam regions, as shown in Fig. 11(b). As observed, the segmentation process effectively eliminates interference from the working platform and background noise.

Compared with the traditional ICP registration algorithm, this study utilizes multi-level voxel downsampling combined with adaptive search distance to improve registration accuracy. Furthermore, restricting correspondence matching to overlapping regions and gradually integrating registered point clouds to increase overlap with subsequent viewpoints further improves registration efficiency and quality.

5. Multi-plane fitting of point cloud

The steel structure workpieces are typically composed of standardized planar plates. Therefore, by performing multi-plane fitting on the complete segmented point cloud of the workpiece, the welding path and torch pose can be calculated based on the obtained plane parameters. To ensure accuracy, robustness, and computational efficiency, an improved RANSAC algorithm is proposed to perform multi-plane fitting on the point cloud in weld seam regions. The process includes multi-plane fitting and parameter optimization.

5.1. Original RANSAC algorithm

The schematic of the original RANSAC algorithm [44] is shown in Fig. 12. Initially, three points are randomly sampled from the original point cloud (blue) to generate a candidate plane. The distances from all remaining points to the fitted plane are calculated, and points within a predefined threshold are classified as inliers (green), while those beyond the threshold are classified as outliers (red). This process is repeated iteratively, and the plane with the maximum number of inliers is selected as the final result.

Specifically, in concrete terms, given three sampled points p_r^1, p_r^2 , and p_r^3 , the plane normal vector $\mathbf{n}_r$ and plane parameter d_r are computed, leading to the plane equation:

$$\mathbf{n}_r = \frac{(p_r^2 - p_r^1) \times (p_r^3 - p_r^1)}{\|(p_r^2 - p_r^1) \times (p_r^3 - p_r^1)\|} \quad (13)$$

$$d_r = -\mathbf{n}_r \cdot p_r^1 \quad (14)$$

$$[\mathbf{n}_r \ d_r] \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0 \quad (15)$$

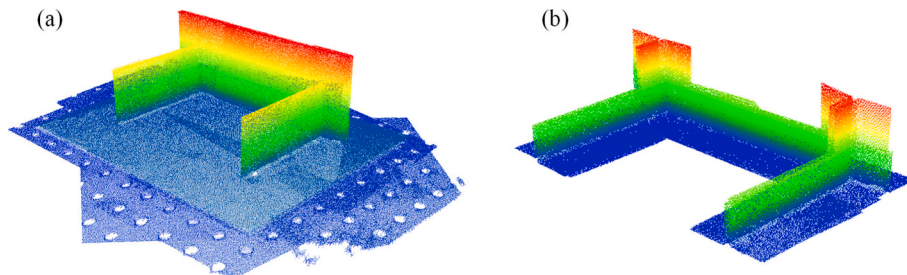


Fig. 11. Fine registered point cloud: (a) unsegmented point cloud; (b) segmented point cloud.

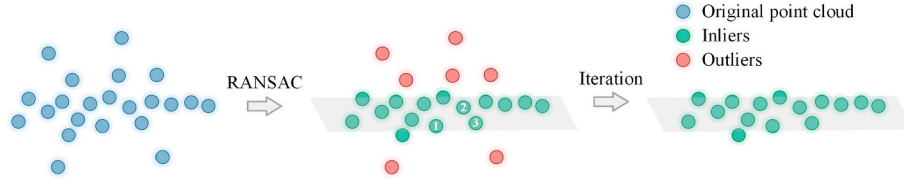


Fig. 12. Principle of the RANSAC algorithm.

Table 3

Distance threshold analysis of plane fitting for the original point cloud.

Distance threshold (mm)	1.0	1.2	1.4	1.6	1.8	2.0	2.2	3.0	4.0	5.0	6.0
Number of failures	42	36	9	9	0	0	0	0	0	18	37

where d_r represents the constant term of the plane equation $a_r x + b_r y + c_r z + d_r = 0$. The point-to-plane distances are then computed as:

$$\tau_2^i = \frac{|a_r x_i + b_r y_i + c_r z_i + d_r|}{\sqrt{a_r^2 + b_r^2 + c_r^2}} \quad (16)$$

The points exhibiting distances below a predefined threshold T_{d2} are classified as inliers. The iterative process proceeds until the maximum iterations limit is attained. Upon completion, the dominant plane is extracted, and its inliers are removed from the point cloud. The remaining point cloud is then processed using the same procedure until all points are included in the fitted plane.

5.2. Parametric analysis of the original RANSAC algorithm

Attributable to the resolution limits of CSL camera and the surface roughness of the workpiece, positional deviations on a single plane approximate ± 1 mm. This observation necessitates the distance threshold T_{d2} in Section 5.1 exceeding 1 mm. Since this parameter is also affected by the point cloud resolution, a KD-tree nearest neighbor search was employed to calculate point-wise distances, quantifying the resolution of the tested workpiece at 0.6427 mm. To determine the optimal parameter, a sensitivity analysis was conducted across various distance thresholds over 50 multi-plane fitting trials. The corresponding segmentation failure rates are listed in Table 3. Given the steel plate thickness of 6 mm, a threshold exceeding this value inevitably leads to the erroneous merging of parallel planes. Consequently, the investigation focused on the range between 1 mm and 6 mm. Experimental results indicate that plane fitting stabilizes when the threshold approximates three times the point cloud resolution. Meanwhile, thresholds approaching the plate thickness still result in merging of adjacent planes, reducing fitting accuracy.

Downsampling serves as an effective strategy to enhance computational efficiency in point cloud processing. To evaluate the impact of downsampling on fitting, different voxel sizes were applied and the minimum thresholds required for 50 successful fits are recorded in

Table 4

Distance threshold analysis of plane fitting for the downsampled point clouds.

Parameters	Original point cloud	Downsampling size of point cloud				
		1	2	3	4	5
Point cloud resolution (mm)	0.6427	0.8661	1.5385	2.4069	3.3599	3.9128
Distance threshold (mm)	1.8	1.9	1.9	1.9	2.0	2.0

Table 4. Due to the low sensitivity of the vertical spacing to down-sampling, the required threshold increases only slightly with decreasing resolution.

Informed by the preceding analysis, a distance threshold T_{d2} of 2 mm is adopted for this study.

5.3. Improvement of the RANSAC algorithm

The original RANSAC algorithm was applied to the complete segmented point cloud, configured with iteration counts of 100 and 500 per plane, respectively. The results in Fig. 13(a) and (b) reveal instances of over-segmentation and excessive fitting of noise planes. Although increasing the iteration count enhances precision, it inevitably incurs a significant computational penalty, prolonging execution time. Furthermore, the original RANSAC algorithm requires manually defined iteration counts, which prevents automatic adaptation to different conditions. To resolve these limitations, the inlier occupancy ratio T_{mpp} and confidence level η_0 are introduced as:

$$\frac{n_q^{\text{in}}}{N_{\text{seg}}} > T_{mpp} \quad (17)$$

$$\left(1 - \left(\frac{3}{N_{fp}}\right)^3\right)^K < 1 - \eta_0 \quad (18)$$

where n_q^{in} is the inlier count for plane q , N_{seg} is the total number of points, and T_{mpp} is the minimum acceptable inlier ratio. N_{fp} is the current subset point count, K is the number of iterations, and η_0 represents the desired confidence level.

The multi-plane fitting process is terminated if the inlier ratio for a plane is below the predefined threshold T_{mpp} . Guided by relevant literature [45], T_{mpp} is set to 0.025, mandating that a valid plane must encompass at least 2.5% of the segmented points. This relatively low threshold was chosen because noise points typically account for only a negligible fraction of the total data. Preliminary experiments demonstrated that this setting is sufficient to retain valid planes while preventing overfitting to random noise.

The confidence level η_0 controls the number of RANSAC iterations. Following standard robust estimation theory [46], η_0 is set to 0.7 in this study. This value ensures a high probability of convergence while avoiding excessive iteration counts. Both parameter configurations were experimentally validated across multiple workpieces with varying surface complexity, confirming their suitability for stable multi-plane detection. With this enhancement, unnecessary planes are eliminated and fitting efficiency is improved. By automatically determining optimal iteration counts, the runtime of plane fitting is significantly reduced. The result of introducing the inlier occupancy ratio is shown in Fig. 13(c), while the cumulative effect of the confidence level is depicted in Fig. 13

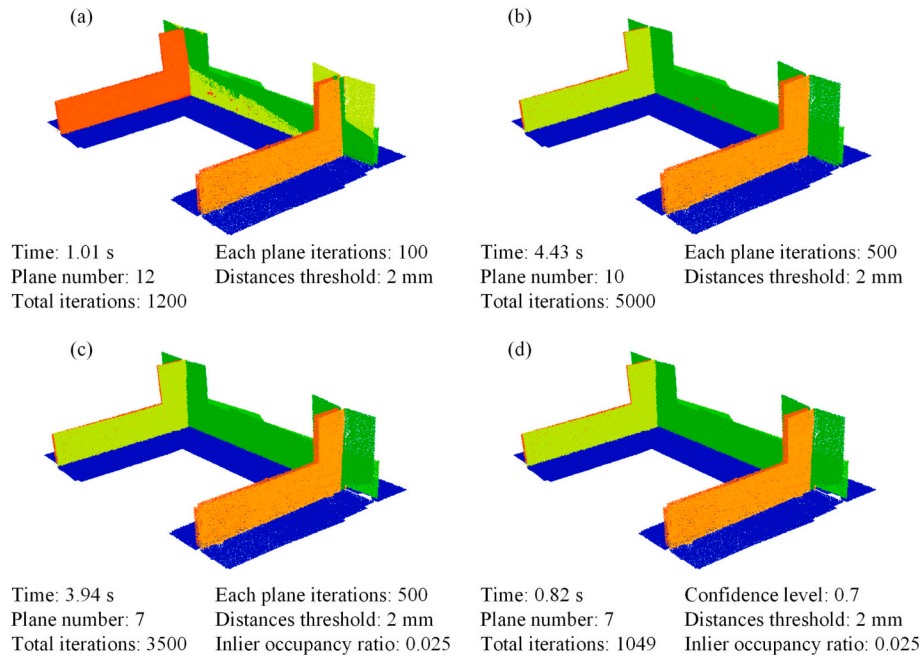


Fig. 13. Comparison of multi-plane fitting results: (a) original RANSAC algorithm (100 iterations per plane); (b) original RANSAC algorithm (500 iterations per plane); (c) introduction of inlier occupancy ratio; (d) introduction of inlier occupancy ratio and confidence level.

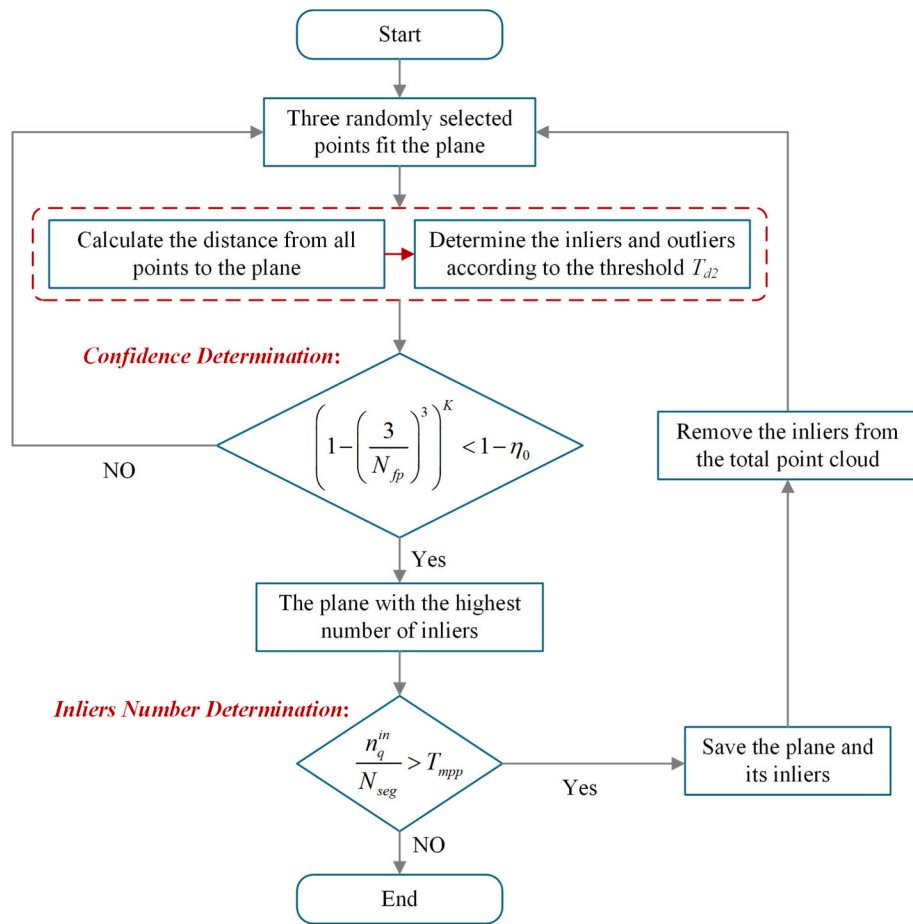


Fig. 14. Flowchart of the improved RANSAC algorithm.

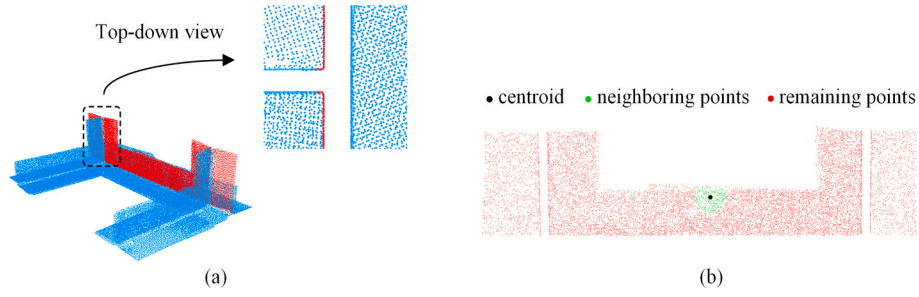


Fig. 15. Optimization of plane parameters: (a) plane over-segmentation phenomenon; (b) acquisition of centroid and its neighboring points.

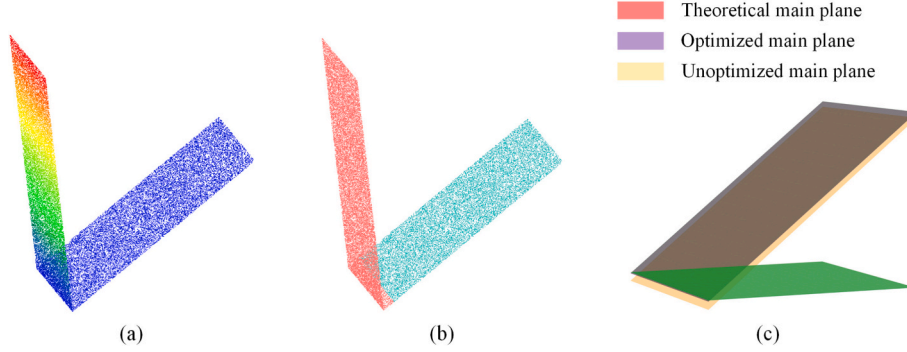


Fig. 16. Comparison before and after plane optimization: (a) original point cloud generated by software; (b) multi-plane fitting; (c) plane optimization result.

(d). The complete algorithmic flowchart is illustrated in Fig. 14.

5.4. Optimization of plane parameters

After multi-plane fitting, independent point clouds are obtained for each plane. However, over-segmentation tends to occur near plane intersections, as demonstrated in Fig. 15(a). This phenomenon arises when points from adjacent planes are incorrectly identified as inliers. To improve fitting precision, a plane optimization method based on centroid analysis is introduced. First, the centroid $\bar{p}_o$ of each plane is calculated. A KD-tree is then constructed to query neighboring points around the centroid, as illustrated in Fig. 15(b). Subsequently, two points (p_o^2 and p_o^3) within this neighborhood are randomly selected for secondary plane fitting. The refined plane parameters are then computed using:

$$\mathbf{n}_o = \overrightarrow{p_o^2 p_o} \times \overrightarrow{p_o^3 p_o} \quad (19)$$

To verify the effectiveness of the proposed plane optimization

method, two theoretical planes were generated in a software environment to simulate the point clouds, as shown in Fig. 16(a). These point clouds were processed utilizing the improved RANSAC-based algorithm. As observed in Fig. 16(b), the main plane (red point cloud) exhibited a noticeable over-segmentation phenomenon. Subsequently, the extracted plane underwent the proposed optimization procedure. The result is presented in Fig. 16(c), where the gray region represents the superposition of the theoretical plane (red) and the optimized plane (purple). Evidently, the optimized plane is virtually coplanar with the theoretical plane, making them visually indistinguishable. In contrast, the unoptimized plane (orange) shows a clear deviation from the theoretical plane. Quantitative analysis indicates that the angular and distance errors for the unoptimized plane were 0.304° and 0.158 mm, respectively. Following optimization, these errors were reduced to 0.011° and 0.004 mm, corresponding to improvements of 96.22% and 97.42%. These findings clearly demonstrate that the proposed plane optimization method effectively mitigates the adverse effects of over-segmentation and significantly enhances the accuracy of plane extraction.

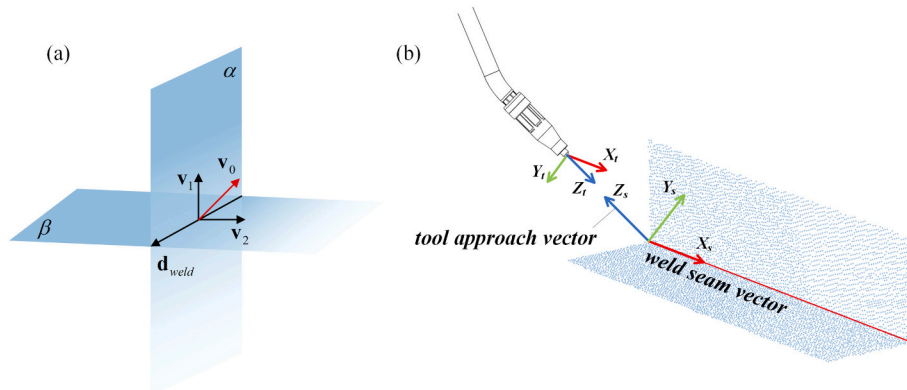


Fig. 17. Welding path generation: (a) plane intersection model; (b) welding path and pose.

6. Generation of welding path and pose

For each pair of orthogonal steel plates, the intersection line represents a fillet weld seam of the workpiece. The geometric relationship between intersecting planes and weld seam enables effective extraction of welding path and pose, as illustrated in Fig. 17(a).

6.1. Calculation of weld seam vectors and tool approach vectors

The two intersecting planes be denoted as $A_1x + B_1y + C_1z + D_1 = 0$ and $A_2x + B_2y + C_2z + D_2 = 0$, with corresponding normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$. The direction vector $\mathbf{d}_{weld}$ of the intersection line is obtained by:

$$\mathbf{n}_1 = [A_1, B_1, C_1], \mathbf{n}_2 = [A_2, B_2, C_2] \quad (20)$$

$$\mathbf{d}_{weld} = \mathbf{n}_1 \times \mathbf{n}_2 \quad (21)$$

A point on the intersection line is computed by solving the system of linear equations under the constraint $z = 0$:

$$\begin{bmatrix} A_1 & B_1 \\ A_2 & B_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -D_1 \\ -D_2 \end{bmatrix} \quad (22)$$

With the direction vector and the known point on the intersection line, the parametric equation of the seam line can be determined. The tool approach vector $\mathbf{v}_0$ is calculated using:

$$\mathbf{v}_0 = \mathbf{v}_1 + \mathbf{v}_2 \quad (23)$$

where $\mathbf{v}_1$ and $\mathbf{v}_2$ are perpendicular to the weld seam direction vector $\mathbf{d}_{weld}$ in two intersecting planes, respectively, as shown in Fig. 17(a).

The extracted weld seam vector and the corresponding tool approach vector are shown in Fig. 17(b). The welding torch pose is then computed based on the relative coordinate transformation between the torch tip coordinate system $X_tY_tZ_t$ and the seam coordinate system $X_sY_sZ_s$, as given in Eq. (24) and illustrated in Fig. 17(b).

$$\begin{cases} X_t = X_s \\ Z_t = -Z_s \\ Y_t = X_t \times Z_t \end{cases} \quad (24)$$

6.2. Determination of weld seam endpoints

To further identify the endpoints of weld seams, intersection points among planes are calculated. Let three planes be $A_{i_p}x + B_{i_p}y + C_{i_p}z + D_{i_p} = 0$, $A_{j_p}x + B_{j_p}y + C_{j_p}z + D_{j_p} = 0$ and $A_{k_p}x + B_{k_p}y + C_{k_p}z + D_{k_p} = 0$, and their common intersection point $(x_{i_pj_pk_p}, y_{i_pj_pk_p}, z_{i_pj_pk_p})$ is computed using:

$$\begin{bmatrix} A_{i_p} & B_{i_p} & C_{i_p} \\ A_{j_p} & B_{j_p} & C_{j_p} \\ A_{k_p} & B_{k_p} & C_{k_p} \end{bmatrix} \begin{bmatrix} x_{i_pj_pk_p} \\ y_{i_pj_pk_p} \\ z_{i_pj_pk_p} \end{bmatrix} = \begin{bmatrix} -D_{i_p} \\ -D_{j_p} \\ -D_{k_p} \end{bmatrix} \quad (25)$$

To avoid invalid intersections caused by adjacent steel plates not being completely parallel, the absolute values of intersection coordinates are restricted to the range $[0.001, 1000]$. The final valid intersection points are marked as green dots in Fig. 18(b). Subsequently, the vertical point clouds are projected onto the weld seam direction:

The unit direction vector of weld seam between plane i_p and plane j_p is $\mathbf{v}_{ij}$, a point in the vertical point cloud be $(x_{i_p}, y_{i_p}, z_{i_p})$, and the reference intersection point be $(x_{i_pj_pk_p}, y_{i_pj_pk_p}, z_{i_pj_pk_p})$. The projection coordinates are calculated using Eqs. (26) to (28), with the principle illustrated in Fig. 18(a).

$$\mathbf{v}_{pi} = (x_{i_p}, y_{i_p}, z_{i_p}) - (x_{i_pj_pk_p}, y_{i_pj_pk_p}, z_{i_pj_pk_p}) \quad (26)$$

$$t_N = \mathbf{v}_{pi} \cdot \mathbf{v}_{ij} \quad (27)$$

$$\mathbf{p}_N = (x_{i_pj_pk_p}, y_{i_pj_pk_p}, z_{i_pj_pk_p}) + t_N \times \mathbf{v}_{ij} \quad (28)$$

where $\cdot$ denotes the dot product and $\times$ denotes the cross product. All vertical point cloud projections onto the weld seam are shown as red points in Fig. 18(b). For a weld seam connected to an intersection point, the endpoints are defined as the intersection point itself and the projection point farthest from it. Conversely, for a seam without an intersection point, the two projection points with the maximum Euclidean distance are identified as the endpoints. These points are highlighted as blue dots in Fig. 18(b).

7. Experimental validation

To validate the effectiveness of the proposed method, a robotic test platform was established, as shown in Fig. 19. The CSL camera is rigidly mounted on the end of the manipulator, forming an “eye-in-hand” system. Using this platform, the proposed framework was applied to identify weld seams and generate paths for four different workpieces.

7.1. Verification of method effectiveness

The tested workpieces have multiple spatially distributed weld seam, with surfaces showing rust and scratches, which are common features in real-world welding scenarios, as shown in Fig. 20. Quantitatively, the rust coverage on workpiece (a) is approximately 80%, while workpieces (b) ~ (d) exhibit coverage ranging from 40% to 60%. Furthermore, all four workpieces show surface scratches with scratch depths within 1 mm.

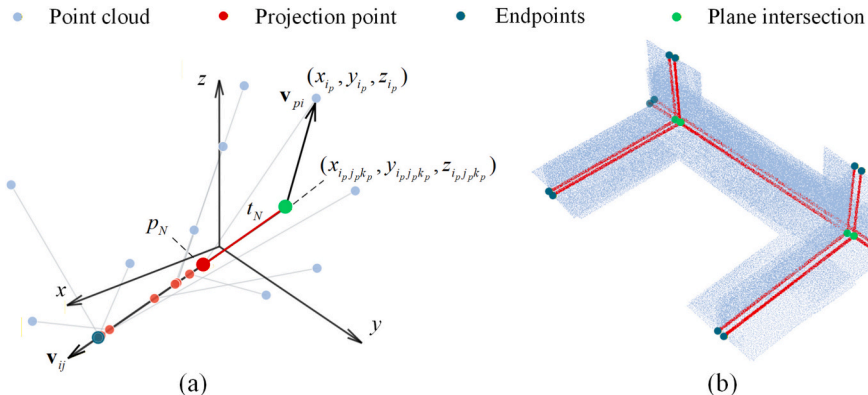


Fig. 18. Point cloud projection and endpoints extraction: (a) point cloud projection; (b) weld seams and endpoints.

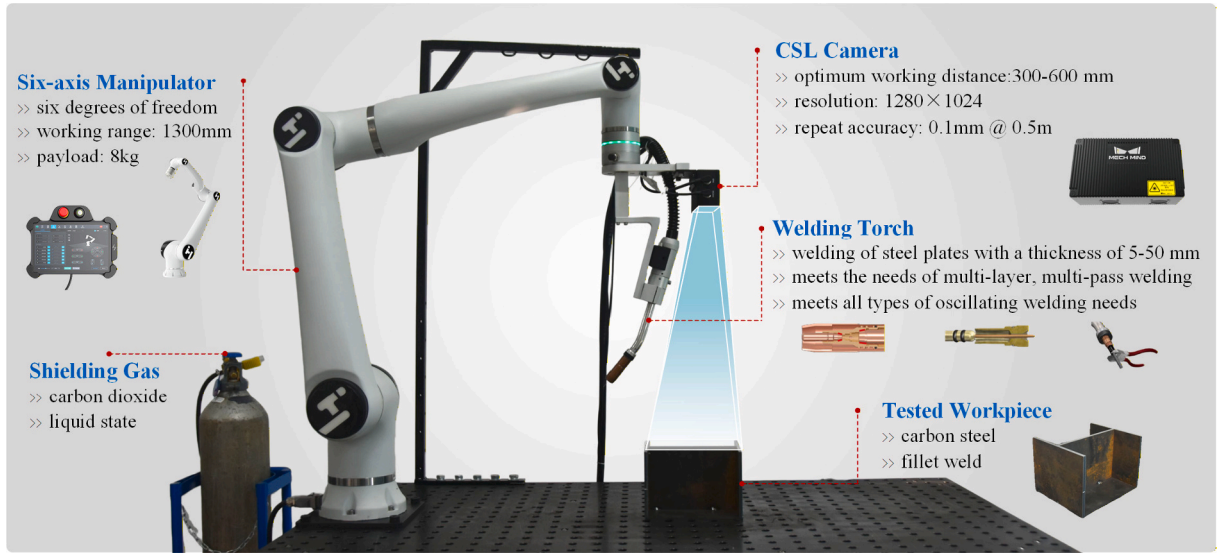


Fig. 19. Robot test platform.

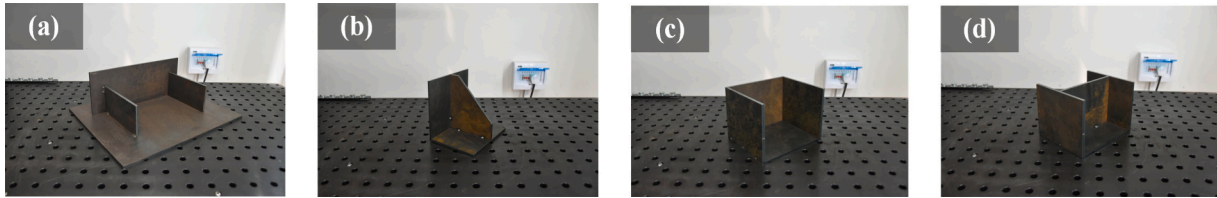


Fig. 20. Tested workpieces.

Table 5

Training hyperparameters of PointNet++.

Parameters	Values	Parameters	Values
Training framework	Pytorch	Learning rate decay	0.5
Epochs	300	Learning rate	0.001
Batch size	16	Point number	4096
Optimizer	Adam	Weight decay	0.0001

7.1.1. Training of point cloud segmentation network

To ensure robust training, a custom-built dataset was constructed comprising single-view point clouds from four distinct workpiece types. This dataset contains various viewpoints and lighting conditions while maintaining data balance. Subsequently, weld seam regions were manually annotated binary labels (“1” for weld, “0” for background). For each tested workpiece, 200 point cloud samples were collected,

resulting in a total of 800 samples.

To validate annotation consistency (intra-rater reliability), 40 samples (10 per workpiece) were re-annotated by the same operator at different times. Comparison of the results revealed a consistency rate exceeded 95%, demonstrating the reliability of the annotations [47,48]. The dataset was subsequently divided into training, validation, and test sets in a ratio of 8:1:1. Meanwhile, 4800 point cloud datasets were generated through data augmentation.

The PointNet++ network was trained using weights pre-trained on the ShapeNet dataset. Training was conducted on an NVIDIA 4060Ti 16GB GPU with the PyTorch framework for 300 epochs. The hyperparameters used for training PointNet++ are listed in Table 5. The final segmentation accuracy and mean intersection over union (mIOU) achieved were 98.16% and 95.07%, respectively, as shown in Fig. 21. During training, the accuracy of the network continued to improve, verifying its convergence and effectiveness. Furthermore, the model

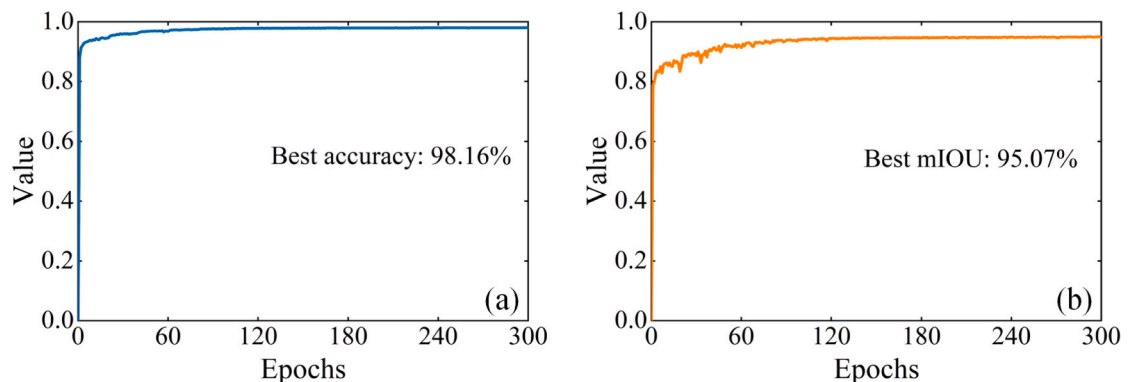


Fig. 21. Training results of PointNet++: (a) accuracy; (b) mIOU.

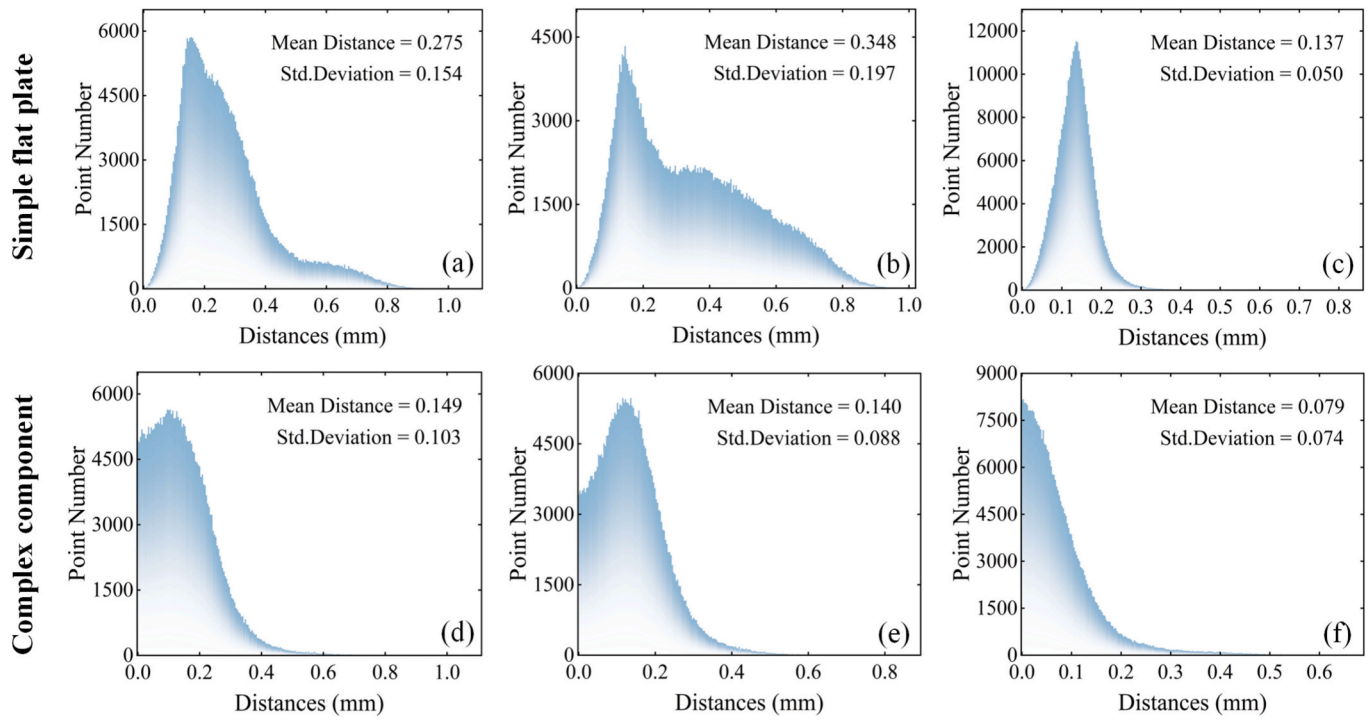


Fig. 22. Comparison of point cloud registration errors: (a, d) coarse registration; (b, e) traditional ICP registration algorithm; (c, f) the proposed algorithm.

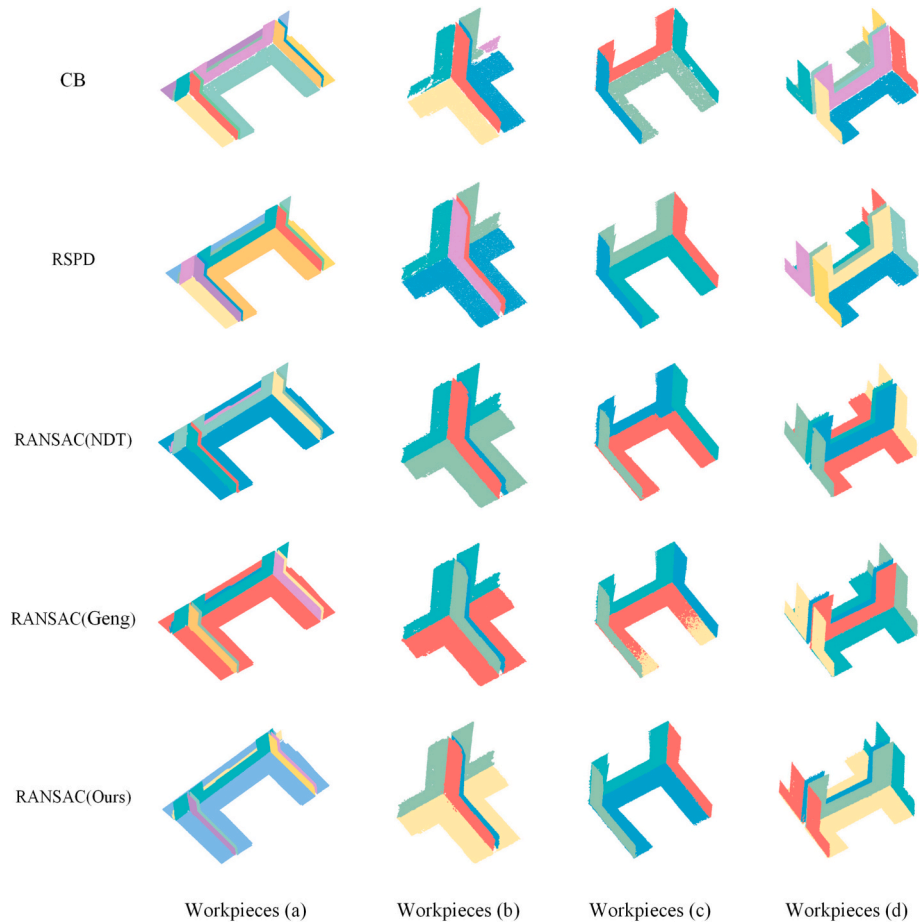


Fig. 23. Performance of different plane fitting methods.

maintained stable performance on unseen samples, demonstrating strong generalization capabilities. These results fully meet the high-precision segmentation requirements for the target workpiece, providing a reliable basis for subsequent weld extraction.

7.1.2. Verification of improved ICP registration algorithm

To conduct a comparative evaluation, point clouds from two adjacent viewpoints were registered using both the traditional ICP algorithm [49] and the proposed method. The test subjects included a simple flat plate and the workpieces(a) in Fig. 20. The initial coarse registration errors are shown in Fig. 22(a) and (d). Subsequently, the registration errors following traditional ICP method are depicted in Fig. 22(b) and (e), while those resulting from the proposed method are illustrated in Fig. 22(c) and (f).

The traditional ICP algorithm exhibits performance limitations across different workpiece types, yielding only limited accuracy improvements for complex workpieces with multiple weld seams while causing a decrease in registration accuracy for the simple flat plate. In contrast, the proposed method consistently enhances registration accuracy for both categories. Specifically, the method achieves a mean matching error is 0.137 mm (std. dev. 0.050 mm) for the flat plates and 0.079 mm (std. dev. 0.074 mm) for the complex workpiece.

7.1.3. Verification of improved RANSAC algorithm

To validate the effectiveness of the improved RANSAC algorithm, comparative experiments were conducted with several representative and state-of-the-art plane detection methods, including the clustering-based (CB) method [50], the robust statistics-based (RSPD) method [51], and two RANSAC methods [40,52]. The CB method detects planes by merging points that are sufficiently close under a smoothness constraint. The RSPD method introduces a split-grow-merge strategy, offering adaptability to varying noise levels. RANSAC (NDT) represents a recent PCA-based solution designed to prevent spurious plane generation, whereas RANSAC (Geng) incorporates a k-nearest neighbor strategy into the RANSAC framework to enhance efficiency. RANSAC (Ours) denotes the improved RANSAC algorithm proposed in this study.

Experiments were performed on original point clouds of four workpieces at full resolution, utilizing the hardware configuration described in Section 2.1. The results of plane fitting are presented in Fig. 23 and Table 6, with the best performance highlighted in bold. As evidenced by the results, the CB, RSPD, and RANSAC (Geng) methods exhibited false detections. The CB method suffered from under-segmentation in plane intersection regions and misclassified valid inliers as noise. The RANSAC (NDT) method manifested obvious over-segmentation and high sensitivity to noise. The RANSAC (Geng) method required frequent manual adjustment of thresholds, leading to reduced practical efficiency. In contrast, the proposed method achieved robust detection of all planes without frequent threshold tuning and demonstrated a distinct superiority in computational efficiency.

Table 6
Details of plane fitting performance.

Method	Workpieces (a)		Workpieces (b)		Workpieces (c)		Workpieces (d)	
	#	T (s)	#	T (s)	#	T (s)	#	T (s)
CB	12/7	105.322	7/4	26.140	4/4	40.325	8/5	45.231
RSPD	12/7	53.691	5/4	13.208	4/4	18.497	8/5	18.269
RANSAC (NDT)	7/7	356.037	4/4	31.152	4/4	72.342	5/5	73.213
RANSAC (Geng)	7/7	1.261	4/4	0.336	5/4	0.680	5/5	0.740
RANSAC (Ours)	7/7	0.187	4/4	0.044	4/4	0.059	5/5	0.075

Number of detected planes over the total number of planes.

7.2. Verification of overall performance

To comprehensively evaluate the proposed framework, this section conducted an assessment of the welding path generation accuracy, efficiency, and applicability. The point clouds of four tested workpieces were captured using a CSL camera, as shown in Fig. 24(a). Applying the registration method described in Section 4, the point clouds from all viewpoints were processed to reconstruct the complete point cloud of the workpieces, as depicted in Fig. 24(b). Subsequently, the rotation matrices obtained from the registration were applied to each single-view segmented point cloud, and the merged point cloud in weld seam region is visualized in the Fig. 24(c). Utilizing the multi-plane fitting method proposed in Section 5, the segmented point clouds were processed, effectively extracting all constituent planes as illustrated in Fig. 24(d). Finally, the welding paths were extracted based on the framework in Section 6, with results shown in Fig. 24(e).

7.2.1. Accuracy verification

To verify the accuracy of the generated welding path, a specific weld seam was selected on each workpiece. The ground truth path was obtained through manual teaching and subsequently compared against the path extracted by the proposed vision-based method. Acknowledging that the teaching process itself introduces errors originating from hand-eye calibration, manipulator TCP calibration, and manual operation, specific experiments were designed to quantify these error sources.

Specifically, for hand-eye calibration, a standard block (30 mm × 9 mm × 5 mm) was employed. Its point cloud was acquired by the CSL camera and transformed according to the hand-eye calibration parameters. The root mean square error (RMSE) of the edge lengths in the transformed point cloud was calculated to quantify the calibration accuracy. For manipulator TCP calibration, the standard four-point method was performed 10 times, with deviations in torch tip coordinates recorded. The translation RMSE was computed to evaluate the TCP calibration precision. Regarding teaching errors, the long edge of the standard block served as the teaching trajectory. This trajectory was executed 10 times, and the first trial was taken as the baseline. The average RMSE of subsequent trajectory relative to this baseline was calculated to represent the teaching error. The RMSE results for hand-eye calibration, TCP calibration, and teaching were quantified as 0.23, 0.36, and 0.28 respectively.

Existing literature [53,54] reports that hand-eye calibration translation errors typically range from 0.2 to 1.0 mm, depending on the calibration plate, number of poses, and pose distribution. Similarly, industrial TCP calibration precision generally falls within 0.1–0.6 mm, while teaching repeatability for well-maintained robots are within 0.05–0.5 mm. Considering the calibration setup used in this study (e.g., number of poses and measurement tools), the quantified errors of the manipulator are consistent with these reported ranges, confirming their reasonableness. Crucially, it must be noted that the cumulative welding path error is a complex function of these factors and is not strictly a linear superposition. In some cases, they may accumulate when aligned in the same direction, and in other cases partial cancellation may occur when errors act in opposite directions. The final path error may be greater than or less than the direct sum of individual error sources. Therefore, the quantified error decomposition in this study provides an estimate of the potential contribution of each factor rather than an exact deterministic accumulation.

Finally, the maximum error (ME) and RMSE of the proposed welding path generation method were calculated, as illustrated in Fig. 25. The maximum ME and RMSE across the four workpieces were 1.16 mm and 0.64 mm, respectively. Benchmarked against relevant studies [37,39–41], the accuracy of the proposed framework falls well within acceptable tolerances, fully satisfying industrial welding requirements.

7.2.2. Performance comparison with existing methods

To comprehensively evaluate the proposed method, a comparative

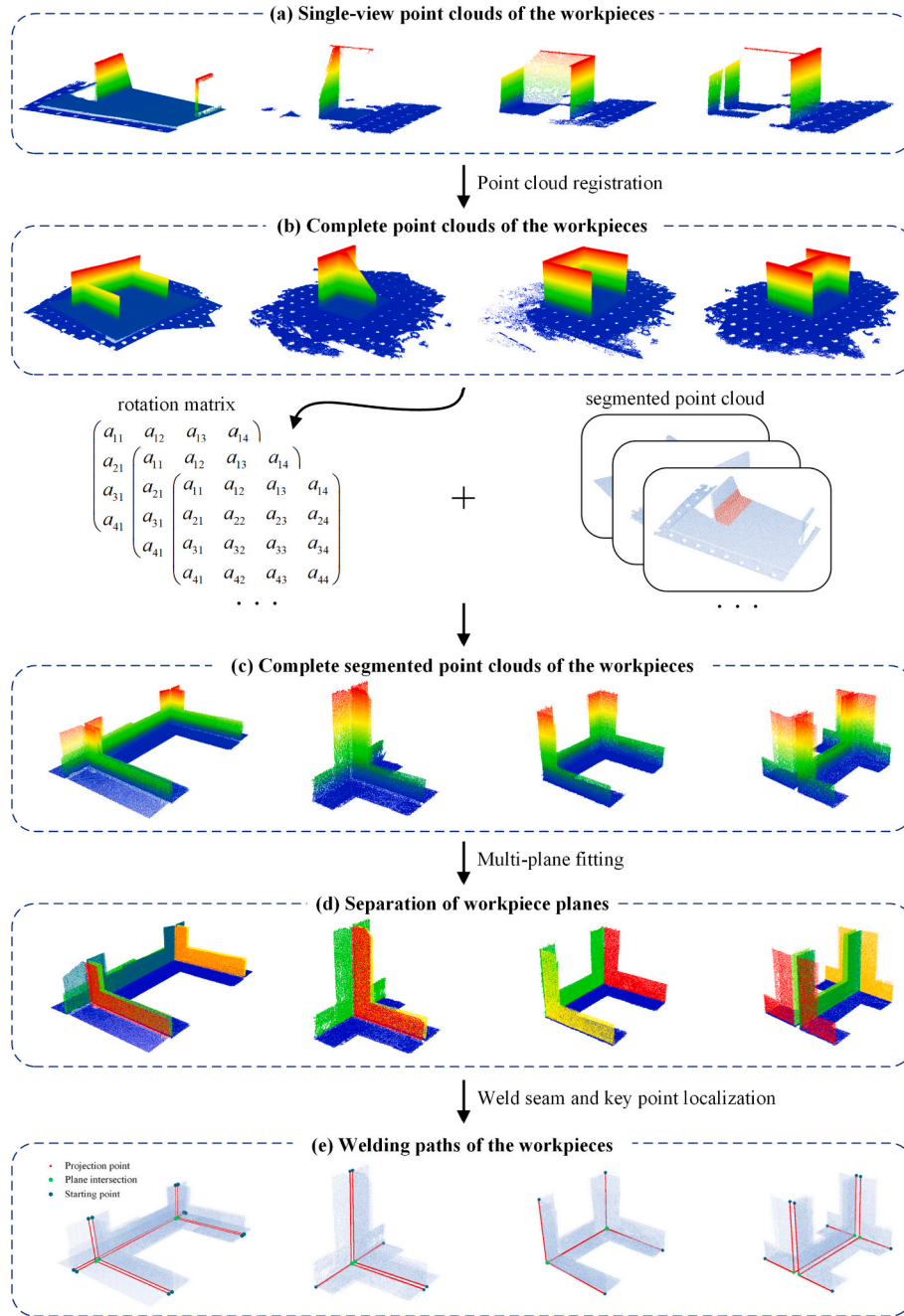


Fig. 24. Test results of different workpieces.

analysis against existing approaches was conducted from multiple perspectives. Given that operational efficiency directly dictates economic viability in industrial production, the runtime performance of the proposed method was prioritized. The breakdown of execution times for each algorithmic step is detailed in Table 7, while a comparison of average runtimes against state-of-the-art methods is presented in Table 8.

It is important to note that the unavailability of source code for the benchmarked methods precludes exact experimental replication regarding hardware configuration and workpiece setup. Nevertheless, according to the descriptions reported in the corresponding literature, the hardware environments and workpiece characteristics (e.g., type and size) are sufficiently comparable to those in this study. Consequently, potential discrepancies are unlikely to introduce significant bias into the efficiency evaluation.

Under such circumstances, methods based on LSL sensors typically achieve the highest processing speed due to their reliance on 2D image processing. However, the inherent requirement for manual scanning trajectory guidance compromises the overall efficiency of weld seam identification. In contrast to existing methods based on CSL sensor and RGB-D camera, the proposed method achieves a significantly reduced average running time (1378 ms) while retaining robust multi-view adaptability.

In other aspects as summarized in Table 8, the proposed framework eliminates the dependency on prior workpiece models (model-free) and accommodates weld seams with various spatial distributions. Through point cloud segmentation, the method effectively mitigates interference from complex environments, ensuring robustness without sacrificing computational efficiency. These results demonstrate that the proposed method offers an effective and practical solution for generating straight

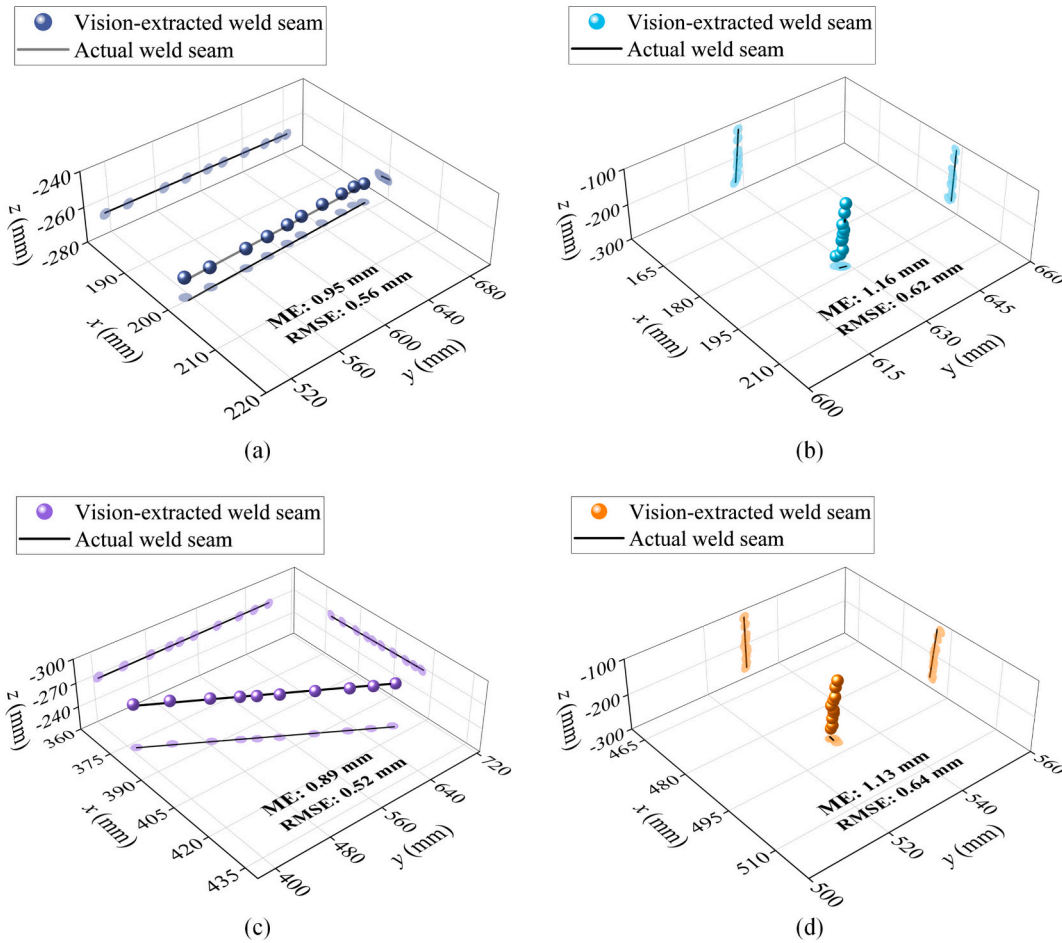


Fig. 25. Comparison of welding path: (a) workpiece 1; (b) workpiece 2; (c) workpiece 3; (d) workpiece 4.

Table 7

Algorithm runtime for different workpieces.

Step	Runing time (ms)			
	Workpiece 1	Workpiece 2	Workpiece 3	Workpiece 4
Point cloud segmentation	1066	1152	1086	1126
Point cloud registration	98	65	94	83
Multi-plane fitting	203	155	87	111
Welding path and pose generation	64	44	35	43
Total time	1431	1416	1302	1363

fillet welding paths, providing a balanced improvement in speed, generalization capabilities, and environmental robustness compared to existing techniques.

Table 8

Comparison of applicability among different methods.

Method	Runing time (ms)	Workpiece model free	Distribution of weld seams	FOV	Adapting to complex environments
Teach-playback	≈ 180,000	Yes	Space	Multi-view	Yes
LSL sensor [23]	57.11	Yes	Space	Single-view	Yes
RGB-D camera [55]	14,090	Yes	Space	Single-view	No
CSL sensors [38]	–	Yes	Flat	Single-view	No
CSL sensors [56]	4715	Yes	Flat	Single-view	No
CSL sensors [40]	3300	Yes	Space	Single-view	No
CSL sensors [41]	2253	No	Space	Single-view	Yes
The proposed method	1378	Yes	Space	Multi-view	Yes

8. Conclusions

To address the challenges posed by complex welding environments and workpiece variability, this paper presented a framework that integrates deep learning with point cloud processing techniques for robust weld seam extraction and automated path generation. The main conclusions are summarized as follows:

- (1) By introducing PointNet++ for weld seam region segmentation, the influence of environmental interference was effectively mitigated. This process filtered out redundant point cloud data, providing a reliable and compact representation for subsequent processing.
- (2) The complete point cloud was reconstructed using a model-free multi-scale registration method. This approach enabled the accurate reconstruction of workpiece geometries without

dependence on predefined models and achieved higher accuracy than traditional ICP algorithm.

- (3) A robust multi-plane fitting method was developed to accurately separate steel plate surfaces. Based on the geometric constraints of intersecting planes, the spatial position of welds and the optimal torch poses were automatically determined.
- (4) Experimental results verified that the generated welding paths satisfy the accuracy requirements of steel structure welding. Furthermore, the combination of efficient path generation method with data reduction achieved through segmentation significantly reduced the overall system runtime. These results indicate that the presented framework achieves a favorable balance among accuracy, efficiency, and adaptability.

Compared to existing weld recognition techniques, which often suffer from sensitivity to environmental noise or high computational costs in unstructured scenarios, the proposed method demonstrates superior robustness and applicability, making it highly suitable for industries such as construction steel structures, mechanical manufacturing, and aerospace. Future research will focus on extending the framework to handle curved weld seams and more complex joint types. Additionally, integrating path optimization with dynamic obstacle avoidance techniques will be explored to further enhance the automation and intelligence of welding robots in complex, dynamic industrial environments.

CRedit authorship contribution statement

Jinxin Yi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xuan Kong:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Hao Tang:** Writing – review & editing, Validation, Software, Investigation. **Jie Zhang:** Writing – review & editing, Visualization, Validation, Software, Investigation. **Zhenming Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Lu Deng:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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Data availability

Data will be made available on request.

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